

Module 1: Overview

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Module Highlights

In this module, we will introduce basic concepts and definitions of design, data description, estimation, inference and prediction. In the first part of the module you will learn the principles of making justifiable inferences and the importance of the random selection of units. In the second part you will learn about design-based inferences and the mechanisms behind permutation and randomization tests. You will learn how to use R to visualize and summarize data, and to perform a randomization test.

 You should plan to complete this module by the end of Week 2.

Module Learning Objectives (MLOs)

Upon completion of this module, you should be able to:

- **MLO1: Describe** the role of statistics in scientific investigations.
- **MLO2: Summarize and display** data using clear statistics and graphics.
- **MLO3: Differentiate** study designs and explain why random selection and assignment matter.
- **MLO4: Apply** the logic of inference with hypothesis tests and randomization methods.
- **MLO5: Interpret** statistical evidence critically, recognizing limitations, misuses, and precision.

Activities & Assignments

In this module, you will:

- **Read** through the Lecture Notes and, optionally, Chapter 1 in the Statistical Sleuth (recommended but not required text).
- **Watch** the Lecture Videos.

- Do the [Lab 01 \(https://canvas.iastate.edu/courses/128073/files/35028361?wrap=1\)](https://canvas.iastate.edu/courses/128073/files/35028361?wrap=1) Activities
- Read through the [R Manual \(https://canvas.iastate.edu/courses/128073/files/35028338?wrap=1\)](https://canvas.iastate.edu/courses/128073/files/35028338?wrap=1) and Work through the [code \(https://canvas.iastate.edu/courses/128073/files/35028360?wrap=1\)](https://canvas.iastate.edu/courses/128073/files/35028360?wrap=1).  [\(https://canvas.iastate.edu/courses/128073/files/35028360/download?download_frd=1\)](https://canvas.iastate.edu/courses/128073/files/35028360/download?download_frd=1) yourself in R.
- Submit [HW01 \(https://canvas.iastate.edu/courses/128073/assignments/2808922\)](https://canvas.iastate.edu/courses/128073/assignments/2808922) and [HW02 \(https://canvas.iastate.edu/courses/128073/assignments/2808923\)](https://canvas.iastate.edu/courses/128073/assignments/2808923)